

Mixed Fujita Freeness and Linear Effective Base-Point Freeness

Kenta Kitamura

9 October 2026 — public draft 2

Abstract

We present proposed proofs of mixed Fujita freeness and the bound $m \geq a + \nu(D) + 1$ for effective base-point freeness on projective klt pairs. The proof establishes numerical freeness criteria by exponential minimization, treats the smooth volume equality case, and descends a nef Cartier divisor to an ample Cartier divisor.

1 Statements

All varieties are integral and projective over $\mathbb{C}$. Divisors and line bundles are written additively. Boundaries have rational coefficients. For a nef Cartier divisor D on an n -fold, its numerical dimension is

$$\nu(D) = \max\{k : D^k \cdot J^{n-k} > 0\},$$

where J is any ample Cartier divisor.

Theorem 1.1 (Mixed Fujita freeness). *Let X be smooth of dimension $n \geq 1$. If $L_1, \dots, L_s$ are ample Cartier divisors, $s \geq n + 1$, and N is nef Cartier, then*

$$K_X + L_1 + \dots + L_s + N$$

is globally generated.

Theorem 1.2 (Linear effective base-point freeness). *Let (X, Δ) be a projective klt pair with $\Delta \geq 0$, and let D be nef Cartier. If $a \geq 1$ is an integer and $aD - (K_X + \Delta)$ is nef and big, then mD is globally generated for every integer*

$$m \geq a + \nu(D) + 1.$$

If $\nu(D) = 0$, then $D \sim 0$.

We prove two numerical statements first.

Theorem 1.3 (Klt numerical freeness). *Let (V, B) be a projective klt pair with $B \geq 0$ and $\dim V = n \geq 1$. Let P be Cartier and $A = P - (K_V + B)$ ample $\mathbb{Q}$ -Cartier. Suppose*

$$A^n > (n + 1)^n, \quad A^d \cdot W \geq (n + 1)^d$$

for every integral proper subvariety $W \subsetneq V$ of dimension $d \geq 1$. Then P is globally generated.

Theorem 1.4 (Smooth numerical freeness). *Let X be smooth of dimension $n \geq 1$, let A be ample Cartier, and let N be nef Cartier. Suppose*

$$A^d \cdot W \geq (n + 1)^d \quad \text{for every integral } W \subseteq X, \quad d = \dim W \geq 1.$$

Then $K_X + A + N$ is globally generated.

2 Valuations and closed threshold loci

Fix a projective klt pair (V, B) with $B \geq 0$, an ample Cartier divisor L , and $n = \dim V$. Write

$$R_j = H^0(V, jL), \quad N_j = \dim R_j = \frac{L^n}{n!} j^n + O(j^{n-1}). \quad (2.1)$$

For a prime divisor F on a smooth model $\rho : T \rightarrow V$, write $K_T + B_T = \rho^*(K_V + B)$ and set $A(F) = 1 - \text{coeff}_F B_T > 0$. We allow nonzero monomial valuations with positive real weights on smooth models adapted to the crepant boundary and all exceptional divisors. If the positive-weight coordinate divisors are E_i and the weights are u_i , then

$$A(v) = \sum_i u_i A(E_i).$$

We use closed centers $c_V(v)$. The value $v(s)$ is the value of the effective zero divisor of s . Both v and $A(v)$ scale by the same positive constant. Rational weights give positive multiples of divisorial valuations by toroidal extraction; real weights are limits of rational weights on the same support.

On a model resolving effective divisors D_i , include their supports, the boundary, and all exceptional divisors in a reduced SNC divisor with components E_j . Here SNC means simple normal crossing. Retraction to the stratum containing the center gives

$$v(D_i) = \sum_j v(E_j) \text{ord}_{E_j}(D_i), \quad A(v) \geq \sum_j v(E_j) A(E_j). \quad (2.2)$$

The first identity is monomialization, and the inequality follows from log canonicity of the reduced SNC divisor on the smooth model. Retraction preserves section values and enlarges the closed center. Weighted orders can be computed at a closed point of the open stratum: the weighted monomial ideals are primary to its defining ideal, so localization and contraction preserve them, as does faithfully flat completion. Unit changes of positive-weight parameters and changes of zero-weight coordinates therefore preserve the order.

Lemma 2.1. *There is $C > 0$ depending only on (V, B, L) such that*

$$v(s) \leq C j A(v) \quad (2.3)$$

for every $j \geq 1$, $0 \neq s \in R_j$, and every allowed valuation.

Proof. Fix a log resolution $\rho : T \rightarrow V$ and a reduced SNC divisor E containing the support of B_T and all exceptional divisors. Choose $0 < \epsilon \leq 1$ with $B_T \leq (1 - \epsilon)E$. Then

$$A(v) = A_T(v) - v(B_T) \geq \epsilon A_T(v).$$

Fix a very ample divisor J on T . A general complete-intersection curve through a closed point z , not contained in the zero divisor of ρ^*s , gives

$$\text{mult}_z(\rho^*s) \leq j(\rho^*L \cdot J^{n-1}).$$

For $n = 1$ this is the degree bound. On a smooth local ring, $\text{lct}_z(g) \geq 1/\text{mult}_z(g)$: by Weierstrass preparation in a general direction, factor the polynomial into its $\text{mult}_z(g)$ roots and apply Hölder's inequality to the one-variable integrals of $|g|^{-2c}$ for $c < 1/\text{mult}_z(g)$. Consequently, for $z \in c_T(v)$,

$$v(s) \leq A_T(v) \text{mult}_z(\rho^*s) \leq \epsilon^{-1}(\rho^*L \cdot J^{n-1}) j A(v).$$

Rational approximation gives the same bound for real weights. $\square$

Lemma 2.2. Fix $x \in V$ and $b > 0$. For all sufficiently large j there is a subspace $U_j \subseteq R_j$ with $\dim U_j \geq \kappa N_j$, for a fixed $\kappa > 0$, such that

$$v(s) \leq \frac{jA(v)}{2b} \quad (0 \neq s \in U_j, x \in c_V(v)).$$

Proof. Choose $0 < \theta < \min\{1/2, 1/(2Cb)\}$ and put $q_j = \lfloor \theta j \rfloor$. For large j , choose $\sigma_j \in R_{j-q_j}$ nonzero at x , and set $U_j = \sigma_j R_{q_j}$. If $x \in c_V(v)$, then $v(\sigma_j) = 0$. Lemma 2.1 bounds every nonzero element of U_j by $Cq_j A(v) \leq jA(v)/(2b)$. Multiplication is injective, and $N_{q_j}/N_j \rightarrow \theta^n > 0$ by (2.1). $\square$

For an ideal $\mathfrak{a}$ we use the pointed threshold

$$\text{lct}_x(V, B; \mathfrak{a}) = \inf_{x \in c_V(F)} \frac{A(F)}{\text{ord}_F(\mathfrak{a})}.$$

A positive finite threshold is computed on a log resolution.

Lemma 2.3. Let $\mathfrak{a}_s$ be a family of nonzero ideals generated algebraically on $V \times S$. For $0 < c < 1$ the locus

$$\{(x, s) : \text{lct}_x(V, B; \mathfrak{a}_s) \leq c\}$$

is Zariski closed.

Proof. Work locally with generators $g_1, \dots, g_q$ and introduce coordinates $u_1, \dots, u_q$ on $\mathbb{A}^q$. Put $G_s = \sum u_i g_{i,s}$. Then

$$\text{lct}_{(x,0)}(V \times \mathbb{A}^q, B \times \mathbb{A}^q; G_s) = \min\{1, \text{lct}_x(V, B; \mathfrak{a}_s)\}. \quad (2.4)$$

Indeed, on a principalization write $g_i = z^\alpha h_i$, with some h_i a unit at each point. The factor $\sum u_i h_i$ is a regular coordinate in the u variables, transverse to the SNC support. The threshold is therefore the minimum of 1 and the ratios on the principalization. The family $G_s = 0$ is a relative effective Cartier divisor, since each fiber ideal is nonzero. Apply semicontinuity of pointed log canonical thresholds for relative effective Cartier divisors on the fixed effective klt pair $V \times \mathbb{A}^q$, and restrict to $u = 0$. Equation (2.4) and $c < 1$ give the assertion. $\square$

Lemma 2.4. Fix j and rational numbers $\alpha_1 \geq \dots \geq \alpha_{N_j} \geq 0$. On $V \times \mathcal{F}_j$, where $\mathcal{F}_j$ is the complete flag variety of R_j , the following condition is closed: there exist a basis (s_i) adapted to the flag and a divisorial valuation v with

$$x \in c_V(v), \quad v(s_i)/A(v) \geq \alpha_i.$$

Its image in V is closed as well.

Proof. Let $\mathfrak{b}_i$ be the evaluation ideal of the first i -dimensional space in the flag. The inequalities are equivalent to $v(\mathfrak{b}_i) \geq \alpha_i A(v)$. Choose an integer $M \geq 2$ such that M/α_i is integral whenever $\alpha_i > 0$, and put

$$\mathfrak{a}_\alpha = \sum_{\alpha_i > 0} \mathfrak{b}_i^{M/\alpha_i}.$$

Existence of v is equivalent to $\text{lct}_x(V, B; \mathfrak{a}_\alpha) \leq 1/M$, since a divisor computes the threshold. Apply Lemma 2.3 to the universal flag ideals. If every $\alpha_i = 0$, the locus is the whole space. Finally use properness of $\mathcal{F}_j$. $\square$

3 Exponential minimization

For $t > 0$ and a basis $\mathbf{s} = (s_i)$ of R_j , set

$$F_{j,t}(v, \mathbf{s}) = \frac{1}{N_j} \sum_i \exp\left(\frac{A(v)}{t} - \frac{v(s_i)}{j}\right), \quad f_{j,t}(x) = \inf_{x \in c_V(v), \mathbf{s}} F_{j,t}(v, \mathbf{s}). \quad (3.1)$$

Rational approximation shows that restricting the infimum to multiples of divisorial valuations does not change it.

Lemma 3.1 (Specialization). *If $x \in W \subseteq V$ and W is integral, then for very general $y \in W$,*

$$f_{j,t}(y) \geq f_{j,t}(x)$$

simultaneously for every j and every $t > 0$ for which $N_j > 0$.

Proof. For each degree and each decreasing rational list in Lemma 2.4, remove its closed realization locus intersected with W if that intersection is proper. There are countably many such loci. A rational list realized at a remaining y is also realized at x . Approximate the normalized values of any test at y from below by such lists. At x rescale the realizing valuations to the same discrepancy as the original test, and pass to the limit in the exponential sum. Taking the infimum proves the claim. $\square$

Fix $x \in V$ and a positive rational number b . The condition needed below is that there exist $0 < t_0 < b$ and $\delta > 0$ such that

$$f_{j,t}(x) \leq 1 - \delta \quad \text{for all sufficiently large } j \text{ and } t \in [t_0, b]. \quad (3.2)$$

Lemma 3.2. *Under (3.2), the infimum in (3.1) is attained for every sufficiently large j and every $t \in [t_0, b]$. There are constants $\kappa > 0$ and $0 < a_* \leq a^* < \infty$, independent of j, t , such that*

$$\kappa \leq f_{j,t}(x) \leq 1 - \delta, \quad a_* \leq A(v) \leq a^*$$

for every minimizing valuation.

Proof. The high-value subspace $\{s : v(s) > jA(v)/(2b)\}$ meets U_j of Lemma 2.2 only in zero, so it has codimension at least κN_j . Every basis thus has at least κN_j elements outside it. Therefore

$$F_{j,t}(v, \mathbf{s}) \geq \kappa e^{A(v)/(2b)}, \quad F_{j,t}(v, \mathbf{s}) \geq e^{-CA(v)}. \quad (3.3)$$

These inequalities bound the discrepancy above and away from zero on the sublevel $F_{j,t} \leq 1 - \delta/2$.

Fix j, t and a minimizing sequence $(v_k, \mathbf{s}_k)$, ordering each basis by decreasing normalized value. After taking a subsequence,

$$A(v_k) \rightarrow a \in [a_*, a^*], \quad v_k(s_{k,i})/A(v_k) \rightarrow q_i.$$

The q_i lie in $[0, Cj]$, and at least one is positive, since otherwise the limiting objective would be $e^{a/t} > 1$. Approximate (q_i) from below by increasing decreasing rational lists. Lemma 2.4 gives nested nonempty closed subsets of the projective flag variety. Choose a common flag and one adapted basis (s_i) .

Resolve its zero divisors and the boundary on one model. Retract the witnesses for the rational lists by (2.2), and normalize their discrepancies to one. On one of finitely many strata the weights belong to the compact simplex

$$u_h \geq 0, \quad \sum_h A(E_h)u_h = 1.$$

Pass to a subsequence and discard zero weights in the limit. The closed center still contains x . Section values are linear in the weights, so the limiting valuation $\tilde{v}$ satisfies $\tilde{v}(s_i) \geq q_i$. The test $a\tilde{v}$ has objective at most the limiting infimum, hence attains it. $\square$

4 Uniform growth and point centers

Lemma 4.1. *Let $\lambda > 0$ be rational. For integral d -dimensional subvarieties W with $L^d \cdot W \geq \lambda^d$, put*

$$I_W(h) = \dim \operatorname{im}(R_h \longrightarrow H^0(W, hL|_W)).$$

There are constants C_d, H_d , independent of W , such that

$$(d!I_W(h))^{1/d} \geq \lambda h - C_d \quad (h \geq H_d). \quad (4.1)$$

Proof. Fix k with kL very ample and $k\lambda$ integral. Set $D_0 = (k\lambda)^d$. In this embedding W has degree $D \geq D_0$. A general linear projection $W \rightarrow \mathbb{P}^d$ is finite of degree D , and its geometric generic fiber consists of D distinct points. Choose D_0 of them. Products of linear forms separating pairs of points interpolate arbitrary values on these D_0 points. Homogenize to degree D_0 with a projection coordinate s_0 .

It follows that the ratios $F/s_0^{D_0}$ of ambient forms of degree D_0 span a space of dimension at least D_0 over $\mathbb{C}(\mathbb{P}^d)$. Choose D_0 independent forms with complex coefficients; rank is unchanged by extending the generic field. Multiply them by all monomials of degree $q - D_0$ in the projection coordinates. Their restrictions to W are independent, so

$$I_W(qk) \geq D_0 \binom{q - D_0 + d}{d}, \quad (d!I_W(qk))^{1/d} \geq k\lambda(q - D_0).$$

For each residue class modulo k , choose a fixed sufficiently large h_r in that class with $h_r L$ globally generated. Multiplication by a section not identically zero on W injects the restriction image in degree qk into that in degree $qk + h_r$. The finitely many h_r give uniform constants in (4.1). $\square$

Proposition 4.2. *Assume (3.2) and*

$$L^d \cdot W \geq \lambda^d, \quad \lambda = (n+1)/b, \quad (4.2)$$

for every positive-dimensional integral proper W containing x . There are $t \in [t_0, b)$ and arbitrarily large degrees m such that every minimizing valuation for $f_{m,t}(x)$ has center $\{x\}$.

Proof. Suppose a minimizing valuation v has center W of dimension $d > 0$. Let Z be its positive-weight stratum on a smooth model, and let r be the number of positive-weight variables. Choose a very general smooth point $y \in W$ satisfying Lemma 3.1, and a point $z \in Z$ over y where $Z \rightarrow W$ is smooth. Avoid additional boundary components. Choose regular parameters at z consisting of r crossing parameters, $n - d - r$ fiber parameters, and d parameters pulled back from affine projection coordinates for a fixed kL embedding of W . Choose the local frame of kL to be the nonvanishing projection coordinate; a compatible formal k th root gives the frame of L .

Write Taylor exponents as $(p, \gamma) \in \mathbb{Z}_{\geq 0}^{n-d} \times \mathbb{Z}_{\geq 0}^d$, and the original weight as $\ell(p)$. Order monomials first by $\ell(p)$, then by $|\gamma|$, and then by an additive monomial well-order. If Γ_j is the set of possible initial exponents of nonzero elements of R_j , elimination and multiplication give

$$\#\Gamma_j = N_j, \quad \Gamma_j + \Gamma_h \subseteq \Gamma_{j+h}. \quad (4.3)$$

Choose bases with these distinct initial exponents. They are adapted to v , and in degree m preserve minimality.

Give tangential parameters weight ε and fiber parameters weight ε^2 . The added coordinate divisors have discrepancy one. Thus

$$A(v_\varepsilon) = A(v) + d\varepsilon + (n - d - r)\varepsilon^2, \quad v_\varepsilon(s) = \ell(p) + \varepsilon|\gamma| + O(\varepsilon^2).$$

For each fixed section the second formula follows by first minimizing the old weight and then the tangential degree. Below any fixed bound there are only finitely many positive-weight exponents; the next old weight has a positive gap. The differentiation is made on a fixed finite basis. The center of v_ε is y . Specialization and minimality make the right derivative of the objective nonnegative. Putting

$$Q_j = \sum_{(p,\gamma) \in \Gamma_j} e^{-\ell(p)/m}, \quad T_m = \frac{\sum_{(p,\gamma) \in \Gamma_m} |\gamma| e^{-\ell(p)/m}}{mQ_m}, \quad T_m \leq \frac{d}{t}, \quad (4.4)$$

we obtain an upper bound for the tangential moment.

For the lower bound, set

$$G_p(j) = \{\gamma : (p, \gamma) \in \Gamma_j\}, \quad b_p(j) = (d! \#G_p(j))^{1/d},$$

with $b_p(j) = 0$ for an empty slice. Restrictions of sections to Z come from W and are independent of the fiber variables. Hence

$$\#G_0(h) \geq I_W(h).$$

The projection coordinates also give $\{0, 1\}^d \subseteq G_0(h_0)$ for $h_0 = kd$. For nonempty finite lattice sets E, F , apply Brunn–Minkowski to $E + [0, 1]^d$ and $F + [0, 1]^d$ to get

$$\#(E + F + \{0, 1\}^d)^{1/d} \geq \#E^{1/d} + \#F^{1/d}.$$

Together with (4.3) and Lemma 4.1, this gives, whenever $G_p(j) \neq \emptyset$ and h is sufficiently large,

$$b_p(j + h) \geq b_p(j) + \lambda h - C'_d, \quad C'_d = C_d + \lambda h_0. \quad (4.5)$$

All constants are independent of the center, the valuation, p , and j . Put

$$B_j = \frac{1}{d!} \sum_p e^{-\ell(p)/m} b_p(j)^{d+1}.$$

For h large enough that $\lambda h - C'_d \geq 0$, convexity yields

$$B_{j+h} \geq B_j + (d+1)(\lambda h - C'_d)Q_j. \quad (4.6)$$

Among subsets of the positive orthant of fixed volume, a simplex minimizes the integral of the sum of coordinates. Apply this to the union of unit cubes at each slice to obtain

$$\sum_{\Gamma_m} e^{-\ell(p)/m} (|\gamma| + d) \geq \frac{d}{d+1} B_m. \quad (4.7)$$

Fix $t \in [t_0, b)$ for the moment and write $f_j = f_{j,t}(x)$ and $c = \liminf_j f_j > 0$. Choose a sequence with $f_m \rightarrow c$. If a positive-dimensional minimizing center occurs infinitely often, pass to a subsequence of a fixed dimension d . Fix an integer $l \geq 2$ and put $j_i = \lfloor mi/l \rfloor$. Summing (4.6) and using (4.7) gives

$$T_m + \frac{d}{m} \geq d \sum_{i=1}^{l-1} \frac{\lambda(j_{i+1} - j_i) - C'_d}{m} \frac{Q_{j_i}}{Q_m}. \quad (4.8)$$

The test $(j/m)v$ in degree j gives

$$\frac{Q_j}{Q_m} \geq \frac{N_j}{N_m} \frac{f_j}{f_m} \exp\left((1-j/m) \frac{A(v)}{t}\right). \quad (4.9)$$

For fixed l , as $m \rightarrow \infty$ we have $N_{j_i}/N_m \rightarrow (i/l)^n$ and $\liminf f_{j_i}/f_m \geq 1$. Using $A(v) \geq a_*$ and $t \leq b$, first let $m \rightarrow \infty$ and then $l \rightarrow \infty$ in (4.8). This yields

$$\liminf_m T_m \geq d\lambda \int_0^1 s^n e^{(1-s)a_*/b} ds. \quad (4.10)$$

Set $\beta = a_*/(b(n+2)) > 0$. Since $e^u \geq 1+u$ and $\lambda b = n+1$,

$$\lambda \int_0^1 s^n e^{(1-s)a_*/b} ds \geq \frac{1+\beta}{b}.$$

Now choose $t \in [t_0, b)$ with $t \geq b(2+\beta)/(2(1+\beta))$. The last bound is strictly greater than $1/t$, contradicting (4.4). The constants a_* and β were fixed uniformly before choosing t . Thus, along the chosen degree sequence, every minimizer eventually has center x . $\square$

5 Rational isolation and lifting

Proposition 5.1. *Keep the preceding notation and assume (3.2) and (4.2). If P is Cartier and*

$$P - (K_V + B) \equiv bL + N$$

for a nef $\mathbb{Q}$ -Cartier divisor N , then P is generated at x .

Proof. Choose $t < b$ and m as in Proposition 4.2. Let $(v, (s_i))$ be a minimizing pair and D_i the zero divisor of s_i . Resolve the D_i , the boundary, and all exceptional divisors on one model $Y_1 \rightarrow V$, with SNC components E_j . Put

$$a_j = A(E_j) > 0, \quad z_j = (\text{ord}_{E_j}(D_i)/a_j)_i.$$

For every stratum Z whose image contains x , form the rational polytope $C_Z = \text{conv}\{z_j : j \in J(Z)\}$. Retraction of the minimizing valuation preserves its objective: a strict decrease of discrepancy would contradict minimality. Thus its normalized profile $q = (v(s_i)/A(v))_i$ belongs to the union of these polytopes. Every point of C_Z is realized by a monomial valuation with normalized discrepancy one, possibly on a larger stratum if some weights vanish.

Write $a = A(v)$. Differentiating first in the scale of v and then in one-sided directions within each polytope containing q , we obtain

$$c_i = \frac{t}{m} \frac{e^{-v(s_i)/m}}{\sum_h e^{-v(s_h)/m}} > 0, \quad c \cdot q = 1, \quad m \sum_i c_i = t < b, \quad (5.1)$$

$$c \cdot z \leq 1 \quad (z \in C_Z, q \in C_Z).$$

The point q belongs to no polytope whose stratum has positive-dimensional image: rescaling its realization to discrepancy a would give a minimizer with a positive-dimensional center.

For each polytope containing q , take its smallest face containing q . Their intersection is rational, contains q in its relative interior, and has $c \cdot z = 1$ throughout. Choose a nearby rational point q^0 in this intersection, avoiding all the finitely many polytopes not containing q . The incidence with all polytopes is preserved. With q^0 fixed, the constraints

$$c^0 \cdot q^0 = 1, \quad c^0 \cdot z \leq 1$$

on the vertices of the polytopes containing q^0 are rational linear constraints satisfied by c . Rational points are dense in the face of the solution polyhedron containing c . We may therefore choose c^0 rational, close to c , with

$$c_i^0 > 0, \quad m \sum_i c_i^0 < b. \quad (5.2)$$

A rational convex representation of q^0 gives a rational monomial ray. Extract it as a divisor F_0 ; its center is x .

Let $I = \{i : q_i^0 > 0\}$, choose $M > 0$ with M/q_i^0 integral for $i \in I$, and set

$$\mathfrak{c} = \sum_{i \in I} \mathcal{O}_V(-(M/q_i^0)D_i).$$

Choose u with $\mathfrak{c}(uL)$ globally generated. On a smooth projective model $\rho : T \rightarrow V$ dominating Y_1 , retain F_0 and principalize $\mathfrak{c}$, with $\mathfrak{c}\mathcal{O}_T = \mathcal{O}_T(-G)$ and all supports SNC. Then $u\rho^*L - G$ is globally generated. For a sufficiently small rational $\eta > 0$, put

$$\Psi = (1 - \eta)\rho^* \left(\sum_i c_i^0 D_i \right) + \frac{\eta}{M} G. \quad (5.3)$$

Writing $A = P - (K_V + B)$, we have

$$\begin{aligned} \rho^* A - \Psi &\equiv \left(b - (1 - \eta)m \sum_i c_i^0 - \frac{\eta u}{M} \right) \rho^* L \\ &\quad + \rho^* N + \frac{\eta}{M} (u\rho^* L - G). \end{aligned} \quad (5.4)$$

By (5.2) this divisor is nef and big for small η .

For each prime component F of the SNC support on T , write

$$a_F = A(F), \quad r_F = \sum_j \text{ord}_F(E_j) a_j \leq a_F.$$

Let S be the reduced sum of those components satisfying

$$x \in \rho(F), \quad r_F = a_F, \quad (\text{ord}_F(D_i)/a_F)_i = q^0. \quad (5.5)$$

It contains F_0 . Each component maps to x : otherwise its retraction would realize q^0 on a stratum with positive-dimensional image. The coefficient of Ψ on every component of S is a_F .

Suppose F meets S . At the image of an intersection point on Y_1 , the crossing components define a common polytope containing q^0 . If $r_F = 0$, then $\Psi_F = 0 < a_F$. Otherwise its retracted profile $q' = (\text{ord}_F(D_i)/r_F)_i$ belongs to that polytope. Set $\mu = \min_{i \in I} q'_i/q_i^0$. Since $c^0 \cdot q^0 = 1$,

$$\mu \leq \sum_{i \in I} c_i^0 q'_i \leq c^0 \cdot q' \leq 1, \quad \frac{\Psi_F}{a_F} = \frac{r_F}{a_F} ((1 - \eta)c^0 \cdot q' + \eta\mu) \leq 1.$$

Equality forces $r_F = a_F$ and $\mu = c^0 \cdot q' = 1$. Since every $c_i^0 > 0$, equality in the weighted average forces $q' = q^0$, including its zero coordinates. The intersection also implies $x \in \rho(F)$, so F is already a component of S . Consequently

$$\Psi_F < a_F \quad (F \not\subseteq S, F \cap S \neq \emptyset). \quad (5.6)$$

Write $K_T + B_T = \rho^*(K_V + B)$ and define

$$\Gamma = B_T + \Psi, \quad Q = \rho^* P - \lfloor \Gamma \rfloor.$$

The divisor Q is integral Cartier and

$$Q - (K_T + \{\Gamma\}) = \rho^* A - \Psi$$

is nef and big. The fractional part $\{\Gamma\}$ is an effective SNC boundary with coefficients less than one. Kawamata–Viehweg vanishing gives $H^1(T, Q) = 0$, hence a surjection

$$H^0(T, Q + S) \longrightarrow H^0(S, (Q + S)|_S). \quad (5.7)$$

Set $C = Q + S - \rho^* P = -[\Gamma] + S$. On S , the coefficient of Γ is $(1 - a_F) + a_F = 1$, so $C_F = 0$. On every other component meeting S , (5.6) gives $\Gamma_F < 1$, hence $C_F \geq 0$. Thus C is effective near S and contains no component of S .

Since S is reduced and all its components map to x , the map $S \rightarrow V$ factors scheme-theoretically through x . Thus $\rho^* P|_S = P_x \otimes \mathcal{O}_S$. Multiply a nonzero element of P_x by the canonical rational section of C , restrict to S , and lift by (5.7). The lift is nonzero at the generic point of every component of S .

The positive part of C is exceptional over V : away from S , a nonexceptional component has $B_T \geq 0$ and $\Psi \geq 0$; on S its coefficient in C is zero. The lifted section is therefore a rational section of P regular in codimension one, and descends by normality. It is nonzero at x , since a section vanishing at x has positive order along every divisor centered at x , whereas the lift has order zero along S . This proves the proposition. $\square$

6 The numerical criteria

Proof of Theorem 1.3. Choose an integer $r \geq 1$ such that $L = rA$ is Cartier, and put $b = 1/r$ and $\lambda = r(n+1)$. Fix $x \in V$. By Lemma 3.1, choose a very general smooth point $y \notin \text{Supp } B$ with

$$f_{j,t}(y) \geq f_{j,t}(x)$$

for all degrees and all $t > 0$. At y use Taylor initial exponents, ordered by total degree and then lexicographically. If H_j is the set of initial exponents, then $\#H_j = N_j$. Filling lattice points in increasing total degree minimizes their sum. From (2.1),

$$\liminf_{j \rightarrow \infty} \frac{1}{jN_j} \sum_{\gamma \in H_j} |\gamma| \geq \frac{n}{n+1} (L^n)^{1/n} > rn = \frac{n}{b}. \quad (6.1)$$

The lattice estimate follows from $\#\{\gamma : |\gamma| \leq R\} = R^n/n! + O(R^{n-1})$ and summation of $|\gamma|$ over this simplex.

Since $A(\text{ord}_y) = n$, the derivative at $\zeta = 0$ of the objective for $v = \zeta \text{ord}_y$ and an adapted basis is

$$\frac{n}{t} - \frac{1}{jN_j} \sum_{\gamma \in H_j} |\gamma|.$$

Choose $t_0 < b$ close enough to b that this is uniformly negative for large j and $t \in [t_0, b]$. Lemma 2.1 bounds the quadratic remainder uniformly. A fixed small $\zeta > 0$ therefore gives $F_{j,t}(\zeta \text{ord}_y, \mathbf{s}) \leq 1 - \delta$ for some fixed $\delta > 0$. Specialization yields (3.2) at x .

For every candidate positive-dimensional center, $L^d \cdot W \geq [r(n+1)]^d$, and $P - (K_V + B) = bL$. Proposition 5.1 applies with $N = 0$. Thus P is generated at every x . $\square$

The smooth case requires the equality case of the lattice estimate.

Lemma 6.1. *Let X, A, N satisfy Theorem 1.4, and put $P = K_X + A + N$. If P is not generated at x , and H_j is the set of Taylor initial exponents at x of $H^0(X, jA)$, ordered by total degree and then lexicographically, then*

$$\liminf_j \frac{1}{jN_j} \sum_{\gamma \in H_j} |\gamma| > n. \quad (6.2)$$

Proof. The lattice estimate gives a lower bound $\frac{n}{n+1}(A^n)^{1/n} \geq n$. If (6.2) fails, then $A^n = (n+1)^n$ and along a subsequence the means tend to n . In that subsequence every fixed open ball with closure in the interior of

$$\Delta = \{u \in \mathbb{R}_{\geq 0}^n : |u| \leq n+1\}$$

meets $j^{-1}H_j$ for all large j . Indeed, omitting such a ball removes a positive multiple of j^n lattice points of total degree at most $(n+1-\epsilon)j$. In a set of N_j distinct lattice points, these must be replaced by points of degree at least $(n+1)j - O(1)$ compared with the set minimizing total degree. The normalized mean would then exceed n by a fixed positive amount.

Choose rational $n/(n+1) < \tau < 1$. The point $c_\tau = (1/\tau, \dots, 1/\tau)$ is interior to Δ . Choose finitely many points of $j^{-1}H_j$, in one sufficiently large degree, whose convex hull contains c_τ in its interior. Intersect this hull with $|u| = n/\tau$, choose rational points whose convex hull contains c_τ in its relative interior, and clear the denominators of their convex combinations. Products of the corresponding sections then give an integer $h > 1$, with $h\tau$ integral, and sections

$$r_\ell \in H^0(X, h\tau A), \quad \text{mult}_x(r_\ell) = nh, \quad (6.3)$$

whose initial exponents β_ℓ satisfy

$$\mathbf{1} \in \text{relint conv}\{\beta_\ell/h\} \quad \text{in the hyperplane } |u| = n.$$

A common multiple makes h as large as needed.

Let $p : Z = \text{Bl}_x X \rightarrow X$ and $E \simeq \mathbb{P}^{n-1}$; for $n = 1$, take p to be the identity and $E = x$. Removing the common order nh from (6.3) gives a subsystem of

$$H = p^*(h\tau A) - nhE.$$

Let $\mathfrak{b}$ be its base ideal. Its restrictions to E are the leading homogeneous forms g_ℓ of degree nh . We claim

$$\text{lct}_z(E; \mathfrak{b}|_E) > 1/h \quad \text{for every } z \in E. \quad (6.4)$$

For the initial monomials, on the chart with the k th homogeneous coordinate nonzero, delete that coordinate from β_ℓ . Projection identifies $|u| = n$ with $\mathbb{R}_{\geq 0}^{n-1}$, so the resulting convex hull contains $\mathbf{1}$ in its interior. Consequently, for every nonzero $w \in \mathbb{R}_{\geq 0}^{n-1}$,

$$\frac{1}{h} \min_\ell \langle w, \beta_\ell^{(k)} \rangle < \sum_i w_i.$$

Retraction to the coordinate divisor shows that the monomial ideal has threshold greater than $1/h$: on its fixed log resolution, all finitely many discrepancy ratios are strictly greater than $1/h$. Choose integer weights realizing the lexicographic initial monomial of each of the finitely many homogeneous forms. Diagonal scaling, with each form divided by its lowest parameter power, gives an algebraic degeneration to these monomials. By Lemma 2.3 and properness of E , some nonzero parameter fiber has threshold greater than $1/h$ everywhere on E . All nonzero fibers are equivalent by diagonal automorphisms of E , proving (6.4). For $n = 1$ the restricted ideal is the unit ideal.

The positivity identity is

$$p^*P - (K_Z + E) - H/h = (1 - \tau)p^*A + p^*N. \quad (6.5)$$

Take a log resolution $\sigma : T \rightarrow Z$ of E and $\mathfrak{b}$, with strict transform E' and $\mathfrak{b}\mathcal{O}_T = \mathcal{O}_T(-G)$. The divisor E' is not a component of G . Put

$$Q = K_T + \sigma^*(p^*P - K_Z - E) - \lfloor G/h \rfloor.$$

This is integral Cartier, and

$$Q - (K_T + \{G/h\}) = (1 - \tau)(p\sigma)^*A + (p\sigma)^*N + (\sigma^*H - G)/h$$

is nef and big. Its last summand is a nonnegative multiple of a globally generated divisor. Thus $H^1(T, Q) = 0$.

Adjunction gives

$$(Q + E')|_{E'} = \sigma_E^*(p^*P|_E) + K_{E'/E} - \lfloor G|_{E'}/h \rfloor. \quad (6.6)$$

Rounding commutes with restriction: distinct SNC components do not restrict to the same prime divisor on E' . By (6.4), the last two terms of (6.6) form an effective divisor. Since $p^*P|_E$ is the trivial bundle with fiber P_x , a nonzero constant times its canonical section lifts to $Q + E'$ by vanishing.

The correction to $(p\sigma)^*P$ is

$$K_{T/Z} - \sigma^*E + E' - \lfloor G/h \rfloor.$$

It has coefficient zero along E' and positive part exceptional over X . As in Proposition 5.1, normality descends the lift to a section of P nonzero at x . This contradicts the hypothesis and proves (6.2). $\square$

Proof of Theorem 1.4. If $P = K_X + A + N$ is not generated at x , Lemma 6.1 and the Taylor expansion used in the proof of Theorem 1.3 give (3.2) for $L = A$, $B = 0$, and $b = 1$. The intersection hypotheses give (4.2) with $\lambda = n + 1$. Proposition 5.1 then generates P at x , a contradiction. $\square$

7 Mixed divisors and Cartier descent

Proof of Theorem 1.1. Put $A = L_1 + \cdots + L_s$. On any integral d -dimensional subvariety W , each mixed intersection number $L_{i_1} \cdots L_{i_d} \cdot W$ is a positive integer. Integrality follows from Cartier intersection; positivity follows by replacing the factors by very ample multiples and taking general complete intersections. Therefore

$$A^d \cdot W = \sum_{i_1, \dots, i_d=1}^s L_{i_1} \cdots L_{i_d} \cdot W \geq s^d \geq (n+1)^d.$$

Apply Theorem 1.4 to A and N . $\square$

Corollary 7.1. *The convex Fujita number of a smooth projective n -fold satisfies $\text{Fu}(X) \leq n + 1$, and this bound is sharp for every $n \geq 1$.*

Proof. Here $\text{Fu}(X)$ is the least nonnegative integer r such that $K_X + \sum_{i=1}^s L_i$ is globally generated for every $s \geq r$ and every collection of ample Cartier divisors L_i . The upper bound follows from Theorem 1.1 with $N = 0$. On $\mathbb{P}^n$, taking all L_i equal to the hyperplane divisor gives $K_{\mathbb{P}^n} + nH = -H$. $\square$

Lemma 7.2 (Cartier descent). *Let D be a Cartier divisor on a normal projective variety X . If qD and $(q+1)D$ are globally generated for some $q \geq 1$, there are a normal projective variety Y , a morphism $f : X \rightarrow Y$ with $f_*\mathcal{O}_X = \mathcal{O}_Y$, and an ample Cartier divisor H on Y such that $D \sim f^*H$.*

Proof. Take the Stein factorization $X \xrightarrow{f} Y \xrightarrow{h} Z$ of the product of the two complete linear-system morphisms. Here Z is the image in the product of projective spaces and h is finite. Let H_q, H_{q+1} be the pullbacks to Y of the two hyperplane bundles. Then

$$f^*H_q \sim qD, \quad f^*H_{q+1} \sim (q+1)D,$$

and $H_q + H_{q+1} = h^*(\mathcal{O}(1,1)|_Z)$ is ample. Set $H = H_{q+1} - H_q$. Then $f^*H \sim D$. The projection formula and $f_*\mathcal{O}_X = \mathcal{O}_Y$ imply that $f^* : \text{Pic}(Y) \rightarrow \text{Pic}(X)$ is injective, so

$$(2q+1)H \sim H_q + H_{q+1}.$$

A line bundle with an ample positive power is ample. $\square$

Proof of Theorem 1.2. The base-point-free theorem for effective klt pairs implies that qD is globally generated for every sufficiently large integer q . Apply Lemma 7.2 to two consecutive such integers to get

$$D \sim f^*H, \quad f_*\mathcal{O}_X = \mathcal{O}_Y,$$

with H ample Cartier and Y normal projective. Put $d = \dim Y$. Intersection numbers of a pullback of an ample divisor give $d = \nu(D)$: the mixed intersections up to degree d are positive, and powers above d vanish. If $d = 0$, then $D \sim 0$.

Suppose $d \geq 1$. Write $R = aD - (K_X + \Delta)$. Kodaira's lemma gives $R \sim_{\mathbb{Q}} A_0 + G$ with A_0 ample and $G \geq 0$. Choose a small rational $0 < \delta < 1$ such that $(X, \Delta + \delta G)$ remains klt. Then

$$R_1 = R - \delta G \equiv (1 - \delta)R + \delta A_0$$

is ample. Choose a small rational $0 < \varepsilon < a$ with $R_1 - \varepsilon D$ ample. For a sufficiently large divisible integer l and a general $J \in |l(R_1 - \varepsilon D)|$, Bertini gives an effective klt boundary

$$\Delta' = \Delta + \delta G + J/l, \quad K_X + \Delta' \sim_{\mathbb{Q}} (a - \varepsilon)D \sim_{\mathbb{Q}} f^*((a - \varepsilon)H).$$

Ambro's descent theorem for an effective projective klt pair with $K_X + \Delta'$ rationally linearly pulled back from a normal proper base gives an effective rational boundary B_Y such that

$$(Y, B_Y) \text{ is klt}, \quad K_Y + B_Y \sim_{\mathbb{Q}} (a - \varepsilon)H. \quad (7.1)$$

Here f has connected fibers and $(a - \varepsilon)H$ is $\mathbb{Q}$ -Cartier. The rank condition is also immediate: on a log resolution $\mu : \tilde{X} \rightarrow X$, write $K_{\tilde{X}} + \Delta_{\tilde{X}} = \mu^*(K_X + \Delta')$. The divisor $\lceil -\Delta_{\tilde{X}} \rceil$ is effective and exceptional, so

$$f_*\mu_*\mathcal{O}_{\tilde{X}}(\lceil -\Delta_{\tilde{X}} \rceil) = f_*\mathcal{O}_X = \mathcal{O}_Y.$$

For any integer $m \geq a + d + 1$, let $P_Y = mH$. By (7.1),

$$A_Y = P_Y - (K_Y + B_Y) \sim_{\mathbb{Q}} (m - a + \varepsilon)H$$

is ample. For every positive-dimensional integral $W \subseteq Y$, with $e = \dim W$, positivity and integrality of $H^e \cdot W$ imply

$$A_Y^e \cdot W = (m - a + \varepsilon)^e H^e \cdot W \geq (d + 1 + \varepsilon)^e > (d + 1)^e.$$

In particular the top-volume inequality is strict. Theorem 1.3 on Y gives global generation of mH . Pulling back gives global generation of mD . This applies to every integer in the asserted range. $\square$